Project Design Phase
Proposed Solution

Date	3 July 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Floods cause significant loss of life and property due to the lack of timely and accurate prediction. Authorities need a reliable system to predict flood risk and support early warning and disaster preparedness.
2.	Idea / Solution description	Develop a Machine Learning-based flood prediction system that uses historical weather and rainfall data to predict flood risk. The system is deployed as a Flask web application where users can enter weather parameters and receive instant flood predictions and safety recommendations.
3.	Novelty / Uniqueness	Uses a trained Decision Tree Machine Learning model (96.55% accuracy) to provide quick and accurate flood predictions through a simple, user-friendly web interface. The solution is lightweight, easy to deploy, and can be extended with live weather APIs in the future.
4.	Social Impact / Customer Satisfaction	Helps disaster management authorities, meteorologists, and communities by providing early flood warnings, improving evacuation planning, reducing disaster impact, and enhancing public safety.
5.	Business Model (Revenue Model)	Government and disaster management agencies can subscribe to the platform. Future revenue opportunities include premium analytics, API integration, cloud deployment, and customized solutions for organizations and research institutions.

6.	Scalability of the Solution	The application can be enhanced with real-time weather APIs, cloud deployment, GIS mapping, IoT sensors, SMS/Email alerts, and support for multiple machine learning models, making it suitable for larger regions and wider adoption.